National University
of Computer and Emerging Sciences
Chiniot-Faisalabad Campus

EE1005 – Digital Logic Design

Assignment Number 1

Fall 2022

Maximum Marks: 45 **Due Date:** 06 September, 2022

Submitted By

Name: ____________________________

Student ID: ____________________________

Section: ____________________________

Submitted To

Dr. Adil Zulfiqar
Assistant Professor & In-charge– EE Department

Submission Date

___ September 2022

Questions	1	2	3	4	5	6	7	8	Total marks
Marks	4	5	2	10	10	4	4	6	45
Obtained Marks									

Question No. 1:

Convert the following numbers with the indicated bases to decimal:

(a) $(4310)_5$ (b) $(198)_{12}$

(c) $(435)_8$ (d) $(345)_6$

Question No. 2:

What is the largest binary number that can be expressed with 16 bits? What are the equivalent decimal and hexadecimal numbers?

Question No. 3:

Convert the hexadecimal number 64CD to binary, and then convert it from binary to octal.

Question No. 4:

Express the following numbers in decimal:

(a) $(10110.0101)_2$ (b) $(16.5)_{16}$

(c) $(26.24)_8$ (d) $(DADA.B)_{16}$

(e) $(1010.1101)_2$

Question No. 5:

Obtain the 1's and 2's complements of the following binary numbers:

(a) 00010000 (b) 00000000

(c) 11011010 (d) 10101010

(e) 10000101 (f) 11111111.

Question No. 6:

Find the 9's and the 10's complement of the following decimal numbers:

(a) 25,478,036 (b) 63, 325, 600

(c) 25,000,000 (d) 00,000,000.

Question No. 7:

Perform subtraction on the given unsigned binary numbers using the 2's complement of the subtrahend. Where the result should be negative, find its 2's complement and affix a minus sign.

(a) 10011 − 10010 (b) 100010 − 100110

(c) 1001 − 110101 (d) 101000 − 10101

Question No. 8:

Represent the decimal number 6,248 in (a) BCD, (b) excess-3 code,